AEP FROM ZERO • PART 1 • KNOW YOUR CUSTOMER (AEP)

EPISODE 02

AEP Architecture: Data Lake vs Profile Store

AEP architecture explained from zero: the five pieces inside Adobe Experience Platform, and the real difference between the data lake and the Real-Time Customer Profile store. Learn what a sandbox is, how to tell a development sandbox from production, and why Enabled for Profile sends one ingest into two worlds and is irreversible for a schema. You will also see what the Edge Network and data governance do, and what real-time actually means in Real-Time CDP, layer by layer. Episode 2 of AEP

Cold open: the menu that looks overwhelming

- Last time: memory, voice, eyes, colleague
- Today: open AEP and name its five pieces
- Data lake or profile store: who keeps what?
- The one switch you can never undo
- What real-time means, layer by layer

KEY IDEA

Yesterday Vikram drew the map. Today he opens Experience Platform itself and shows Kavya that its busy menu is really just five pieces.

Five pieces, one mental model

Piece	Think of it as	Its job
Sandbox	A practice kitchen	Keeps work isolated
Datasets and data lake	The archive	Keeps complete history
Profile store	Receptionist, perfect memory	Knows each person now
Edge Network	The outpost	Decides near the customer
Governance	Legal department in the pipes	Stops misuse of data

KEY IDEA

Sandboxes, the data lake, the profile store, the Edge Network and governance. Learn these five and the menu stops being scary.

What is a sandbox in AEP?

Sandbox	Development	Production
What it holds	Practice and test data	Real customer data
How to treat it	Play freely, reset it	Treat it with respect
Your habit	Build and break here	Look, screenshot, leave

KEY IDEA

A sandbox is a complete, isolated copy of the platform. Development sandboxes are practice kitchens; production holds real customers.

Which kitchen am I in? The sandbox switcher



KEY IDEA

As of 2026 the sandbox switcher sits just under the top bar and shows the current sandbox and its type: Development or Production.

Datasets and the data lake: the archive

- Dataset: a managed home for one kind of data
- Every dataset follows a schema, its blueprint
- All datasets together: the data lake
- Keeps years of history, row by row
- Used for queries, exports and analysis
- Thorough, but not fast

KEY IDEA

Every piece of data lands in a dataset, and all datasets together form the data lake: thorough, complete history, not fast.

The profile store: the receptionist

- One merged, current view per person
- Attributes plus recent events
- Answers in milliseconds
- “Is Priya Gold right now?” Ask here
- Journeys read the profile, not the lake

KEY IDEA

The Real-Time Customer Profile keeps one merged, current view of each person and answers in milliseconds. Journeys ask it, not the lake.

Data lake vs profile store: the difference

Compare	Data lake	Profile store
What it keeps	Complete history	The current person
Speed	Thorough, not fast	Milliseconds
Who asks it	Queries, exports, analysis	Journeys and audiences
Think of it as	The archive	The receptionist

KEY IDEA

The data lake keeps complete history and is thorough but slow. The profile store keeps each current person and answers in milliseconds.

One ingest, two worlds

A row arrives, say a new order

▼

It lands in its dataset, in the lake

▼

Is the dataset Profile-enabled?

▼

No: it stays archive-only

▼

Yes: also folded into the profile

▼

A journey reads the profile

KEY IDEA

A Profile-enabled dataset sends each row into both worlds: the lake keeps the history, the profile store keeps the current person.

The switch you can never undo

- Schema Profile switch: permanent once saved
- Dialog: “the schema can't be disabled”
- Needs a primary identity first (episode 4)
- Design and check on paper, then enable
- Never enable a schema that is not yours

KEY IDEA

Enabling a schema for Profile is irreversible. As of 2026 the dialog warns that once enabled and saved, the schema can't be disabled.

The Edge Network: the outpost

- Adobe servers physically close to customers
- A light copy of what instant decisions need
- For choices inside the same page load
- Example: a banner chosen as the page loads
- Bloom sends events through it in episode 12

KEY IDEA

Some decisions cannot wait for a trip to a data centre. The Edge Network holds what instant decisions need, close to the customer.

Governance: the legal department in the pipes



KEY IDEA

Labels on fields say what data is and how it may be used; policies enforce them automatically, so the platform itself blocks misuse.

What does real-time actually mean?

Layer	How fast it really is
Profile lookup	Milliseconds
Edge decision	Within the same page load
Streaming audience	Near real time, no minutes figure
Batch audience	Once every 24 hours

KEY IDEA

Real-time means different speeds at different layers: milliseconds for a lookup, one page load at the edge, near real time for streaming, daily for batch.

Real-time at Bloom: match the speed to the job

Bloom job	Speed it needs
Is Priya Gold right now?	Profile lookup: milliseconds
A banner while the page loads	Edge: same page load
Welcome a brand-new customer	Streaming: near real time
Win back the 90-day lapsed	Batch: every 24 hours is fine

KEY IDEA

Faster is not always better. Match each Bloom job to the speed it truly needs, by asking how quickly the business must react.

Interview corner: one ingest, two worlds

- One ingest, two worlds
- Lake: history. Profile store: current person
- A journey asks the profile store
- Real-time? First ask: which layer?
- Schema Profile switch: design first

KEY IDEA

Say it in one sentence: one ingest, two worlds. The lake keeps the history, the profile store keeps the current person.

Recap, your question, and what comes next

- Five pieces: sandbox, lake, profile, edge, governance
- One ingest, two worlds; schema switch is permanent
- Real-time depends on the layer
- Your question: lake or profile store?
- Next: XDM, the shipping container

KEY IDEA

Five pieces, one ingest two worlds, and real-time by layer. Next: before any data arrives, Bloom must agree what a customer is.